

Malek Ahmad | majobs112002@gmail.com | +972549498961 | [Linkedin](#) | [Github](#) | [Portfolio](#)

Software Engineer specializing in **C/C++** and **Python**, with strong proficiency in **system-level programming**, **performance-oriented development**, and **Linux-based environments**. Strong foundation in **machine learning and AI**, combining analytical thinking with practical system design. A **fast and self-driven learner** with **strong problem-solving skills**, **highly motivated team player**, and able to **work effectively under pressure**.

Skills

Programming languages: C, C++, Python, JavaScript.

IDEs: CLion, PyCharm, VS Code, Cursor.

Operating Systems: Linux (processes, memory, system calls), Windows.

Development Tools: Git, pytest, Valgrind.

System Programming: TCP/IP Socket Programming, Parallel Programming (MPI, OpenMP), Multithreading.

Database: SQL (MySQL), Firestore, MongoDB, SQLite.

Education

2021 –2025: Azrieli College of engineering | Software engineering | GPA 90.83

Courses: C/C++ Programming | Operating Systems | Computer communication | Data Structures | Algorithms | Object Oriented Programming | Parallel and distributed computation | Reinforcement Learning

Projects

- **2026: AI Codebase Assistant**, [Code link](#)

Built an **AI-powered codebase assistant** for analyzing GitHub repositories and answering repository-specific questions using natural language.

Developed a full-stack solution using **Python (FastAPI)**, **React**, **OpenAI**, and **PageIndex**, implementing **tree-based retrieval**, alongside **PDF export**, and **Google Drive integration**.

- **2025: Retrieval-Augmented Generation (RAG) System for Multilingual Document Search**, [Code link](#)

Built a **multilingual document-search system** using **Flask**, **LangChain**, and **FAISS**, enabling **accurate Q&A** over large document collections.

Designed **backend architecture**, **vector indexing pipeline** and **REST APIs**, focusing on modularity and efficient data processing.

- **2025: Parallel Image Matching System**, [Code link](#)

Implemented a **hybrid parallel image-matching system** in **C** using **MPI** for distributed multi-process execution and **OpenMP** for intra-process thread parallelism, with **dynamic workload** distribution across cluster nodes.

- **2024: Indexed File System Simulator**, [Code link](#)

Implemented a **block-based file system simulator** modeling inode structures with **direct**, **single-indirect**, and **double-indirect allocation** using **C++**.

Designed **manual block management** using a **BitVector** and performed **low-level disk I/O operations** to simulate real storage behaviour.

Certificates

- **2026: AI Growth Lab Program — Place-IL**, [Certificate link](#)

- Completed a **competitive generative AI program** focused on modern **LLM applications**, **full-stack AI development**, and **hands-on implementation of real-world AI products**.

- **2025: Place-IL Developers Internship Program**

- Successfully passed a **competitive selection process (250+ candidates)**, including **coding challenges** in **OOP**, **data structures**, and **databases**.
- Successfully completed **HR**, **technical**, and **group** interview stages.

Experience

2022 – 2025: Tutor | Azrieli College of Engineering | Jerusalem

Tutored **4-5 students per year** in **programming**, **algorithms**, **math** and **physics** courses, improving my responsibility, self-confidence, and socialization skills.

Languages

Arabic (Native) | Hebrew (Fluent) | English (High level)